

● **PRODUCT #1 BUILD GUIDE**

Earn / Morpho Yield & Risk Intelligence

Vault Explorer · DeFi Pools Repo · Portfolio Tracker
How to Build It & Keep Data Live

Euler Explore–style registry for Morpho vaults, Robinhood Earn, LPs & pools
on Robinhood Chain — with architecture, real-time updates, costs, and roadmap

What you are building

A SaaS app that (1) explores Morpho vaults and yield pools like Euler Finance Explore, (2) explains Robinhood Earn risk and APY truth (base vs incentives), (3) tracks wallet portfolio positions, and (4) later expands to Uniswap LPs and other adapters — without custody and without requiring a full DeFiLlama rebuild on day one.

Reference UX: <https://app.euler.finance/explore?network=1>

Chain focus: Robinhood Chain (chain ID 4663) · Morpho · USDG Earn

Disclaimer: Not financial, legal, or investment advice. Yields are variable. Geo restrictions apply.

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1. Product Vision & Positioning

Robinhood Chain's Earn product routes USDG into a Morpho vault (curated by Steakhouse). Retail users see a simple “~7% APY” story; underneath, rates are variable, partly incentives, with smart-contract, liquidity, curator, and stablecoin risks. Your product is the **intelligence and explore layer** — not a bank, not a new lending protocol.

Layer	Euler-like	Your version
Explore	Browse vaults/markets	Morpho vaults, USDG Earn, later Uniswap LPs
Detail	One vault APY/TVL/risk	Net APY vs incentives, utilization, curator, liquidity
Portfolio	Your positions	Wallet balances, earned yield, alerts

Table: Three layers of the product shell.

You do **not** need to re-index all of DeFi on day one. Start narrow (Morpho + Robinhood Earn on RH Chain), then widen with adapters. Euler wins because it is deep on Euler vaults — not because it lists every farm on earth.

2. Scope Phases (MVP → Full Repo)

Phase 1 — MVP (4–6 weeks): Robinhood Chain + Morpho only

- Explore table: vault name, asset (USDG...), TVL, supply APY, net APY, rewards APY, curator, chain
- Vault detail: market allocations, historical APY chart, utilization / withdraw liquidity, fees, risk notes
- **Robinhood Earn spotlight** card (Steakhouse-curated vault users actually use)
- Connect wallet → my deposits in those vaults
- Simple alerts (APY drop, high utilization)

Phase 2 — Portfolio tracker

- All positions across vaults
- Cost basis / earned yield estimate
- CSV export
- Multi-wallet later

Phase 3 — “Repo of everything”

- Uniswap V3/V4 LP positions on Robinhood Chain
- Other lending markets as they appear on RH Chain
- Optional: Ethereum/Base Morpho for comparison (multi-network like Euler)

Scope rule

Do not try to be DefiLlama + Euler + Zapper in month one. Depth on Morpho + Earn clarity is the wedge. Breadth (every LP) is a later growth feature.

3. How Euler-Style Apps Get Data

Apps like Euler Explore do not scrape their own frontend. They combine onchain truth with hosted indexes:

Onchain contracts → Indexer / Hosted API → Your backend cache → Frontend (truth) (Morpho API / subgraph) (Postgres) (Explore table)

Source	Role	Freshness
Protocol GraphQL API (Morpho public API)	Vault lists, APY, net APY, rewards, TVL, allocations, positions	Seconds–minutes (pre-indexed)
Subgraph (The Graph / Goldsky)	Events: deposits, withdraws, factory deploys	Near real-time on new blocks
RPC + viem readContract	Live balances, share price, allowance for connected wallet	Live per request or poll
Your database	History, favorites, alerts, curated labels (“Robinhood Earn”)	You control
WebSocket / new block heads	Optional push when a new block lands	True near real-time

Table: Data source stack for yield explorers.

Morpho specifically publishes a public Morpho API (GraphQL) so you can skip building a full indexer at first: vaults, APYs (native + rewards), positions, markets. Same idea as using Euler’s API/subgraph instead of reading every storage slot yourself. Prefer API first; add custom indexing only when RH Chain coverage is incomplete or you need custom history.

4. How Real-Time Updates Work

Nothing in retail DeFi UIs is literally continuous streaming for every table row. You combine three loops:

A. Server refresh (catalog of vaults)

Cron or worker every **15–60 seconds**:

- Query Morpho API: vaults on chainId **4663** (Robinhood Chain); optional Ethereum for comparison
- Normalize fields: address, symbol, asset, tvlUsd, apy, netApy, netApyExcludingRewards, rewards[], curator, fees
- Upsert into Postgres
- Frontend reads **your** API (fast, stable, cacheable — browsers never hammer Morpho directly)

This is how Explore stays “live” without rate-limit meltdown.

B. User portfolio (more live)

When a wallet is connected:

- **On page load**: balanceOf, vault convertToAssets, market positions via API or contracts
- **Poll every 10–30s** while tab is open (SWR / React Query refetchInterval)
- **On new block (optional)**: Alchemy/WebSocket newHeads → invalidate queries
- **On user tx**: after deposit/withdraw receipt → immediate refetch

C. What is never real-time without extra work

- Historical charts → stored snapshots (cron writes APY/TVL every 5–15 min)
- Incentives / Merkl-style rewards → Morpho rewards fields and/or reward APIs
- Labels (“Robinhood Earn”, “deprecated”, “verified”) → your curated tables (like Euler known-vault metadata)

Update interval cheat sheet

Data	Update method	Interval
Vault list + TVL + APY	Backend poll Morpho API	30–60s
Wallet shares / positions	Frontend + RPC (and/or Morpho positions API)	10–15s or on block
Charts / history	DB snapshots	5–15 min
Alerts	Backend compare new APY vs threshold	Same as catalog poll

Data	Update method	Interval
Prices (ETH, USDG)	Chainlink / price API	15–60s

Table: Practical freshness targets for MVP.

5. System Architecture

██ ■ Next.js App (Explore | Vault | Portfolio
| Alerts) ■ ■ wagmi/viem + wallet connect + SWR/React Query ■
██ ■ REST / tRPC
██ ▼ ██ ■ API (Node / Next route handlers) ■ ■ GET
/vaults?chain=4663 · GET /vaults/:address ■ ■ GET /portfolio/:wallet · POST /alerts ■
██ ■ ■ ▼ Postgres + Redis Morpho GraphQL
API (cache, history, labels) + RH RPC (Alchemy) ▲ ■ Worker (every 30s) morpho sync → upsert vaults → evaluate alerts

Browsers talk only to your API. Workers own external protocol APIs. RPC is used for wallet-specific truth and write previews — not for scanning all vaults every second from the client.

6. Recommended Tech Stack

Piece	Choice	Why
Frontend	Next.js App Router + Tailwind	Fast ship; Euler-like tables/filters
Chain reads	viem	Modern, typed; RH Chain EVM
Wallet	wagmi or Privy	wagmi simple; Privy if social login later
Catalog data	Morpho public GraphQL API first	Skip full indexer initially
RPC	Alchemy Robinhood mainnet	Recommended by RH docs; AA later
DB	Postgres (Supabase / Neon)	Vaults, snapshots, alerts
Cache	Redis (optional early)	Portfolio keys, rate-limit protection
Jobs	node-cron, Inngest, or Railway worker	30s vault sync
Auth (optional)	Privy / Clerk / NextAuth	Only for saved alerts & Pro
Payments	Stripe	Pro tier \$10–20/mo
Hosting	Vercel (FE) + Railway (worker/API)	Common indie pattern

Table: Stack choices aligned with RH Chain and solo/small-team speed.

7. Data Model

```
chains(id, name, chain_id) vaults( address, chain_id, name, symbol, asset_address, asset_symbol, curator, tvl_usd, apy,
net_apy, net_apy_ex_rewards, rewards_apy, performance_fee, management_fee, is_verified, tags[], updated_at )
vault_snapshots(vault_id, tvl_usd, net_apy, ts) -- charts vault_labels(vault_id, label) -- robinhood-earn, steakhouse
user_watchlists(user_id, vault_id) alerts(user_id, vault_id, type, threshold, channel, active)
```

Curated seed (critical for positioning)

Tag the official **Robinhood Earn / Steakhouse Morpho vault** as robinhood-earn. Show incentive vs base APY clearly. That split is your product edge over a raw Morpho UI or a generic yield list.

8. Explore UX (Euler-like)

Copy the **interaction pattern** of Euler Explore, not the brand. Reference: <https://app.euler.finance/explore?network=1>

- **Network selector** — start with Robinhood Chain (4663); optional Ethereum later
- **Filters** — asset (USDG), curator, min TVL, has rewards, verified only
- **Sort** — net APY, TVL, name, recently updated
- **Columns** — Asset | Vault | TVL | Base APY | Rewards | Net APY | Curator | Risk tag
- **Row click** → vault detail page
- **Connect wallet** → “Your deposits” column and/or Portfolio tab

Mobile: sticky filters collapsed; horizontal scroll on table or card list. Empty states when RH Morpho listing is thin — show “verified Earn vault” first.

9. Portfolio Tracker Design

User connects wallet → API: Morpho positions-by-user (if available for chain) → OR loop known vault addresses: `convertToAssets(balanceOf(user))` → Cache ~15s in Redis keyed by wallet+chain → UI polls every 15s + refetch on window focus + refetch after tx

PnL (good enough v1)

- Earned $\approx$ current assets – net deposits (from Deposit/Withdraw events or share delta over time)
- Store first-seen balance snapshot when user links wallet
- Do not promise tax-grade accuracy until full historical indexing exists
- CSV export of positions + timestamps for accountants (Pro feature)

10. Risk Intelligence Features (Your Edge)

A plain vault table is a commodity. Your SaaS sells **clarity**:

Feature	What user learns	Data source
Base vs rewards APY	How much of 7% is temporary incentives	netApY vs netApYExcludingRewards
Utilization / liquidity	Can I withdraw right now?	Morpho market/vault state
Fee drag	What curator/protocol takes	performanceFee, managementFee
Curator label	Who sets risk parameters (e.g. Steakhouse)	API + your labels
Incentive end scenarios	What if rewards go to zero?	Computed from fields above
Plain-language risks	Smart contract, peg, liquidity, curator	Your content + RH disclosures
Alerts	Notify when APY drops or util spikes	Your worker + user settings

Table: Differentiated risk intelligence features.

Never market yields as guaranteed. Never imply FDIC/SIPC. Robinhood’s insurance (Lloyd’s/RELM) is not a personal user policy and does not cover market/lending losses — reflect that honestly.

11. Week-by-Week Build Plan

Week	Ship
1	Next.js shell, RH Chain wallet connect, hardcoded 1–3 vault addresses (Earn vault), live APY/TVL via Morpho API or contracts
2	Explore table from API sync worker + Postgres; vault detail page; charts from snapshots
3	Portfolio: wallet positions; deep-link deposit/withdraw to Morpho or Robinhood; no custody

Week	Ship
4	Alerts (email/Telegram); Robinhood Earn explainer module; Pro feature flag
5–6	Polish filters, mobile, SEO landing; optional Uniswap LP positions; Stripe \$10–15/mo

Table: 4–6 week path to a chargeable MVP.

12. Real-Time Levels (Complexity Ladder)

Level	What users feel	How	When
1 — Good enough	Updates every ~30s	Cron + SWR poll	Start here
2 — Snappy	Wallet updates every block	WebSocket newHeads + invalidate	After MVP
3 — Pro	Live activity feed	Index Deposit/Withdraw events + WS to clients	Growth
4 — Full Zapper	Every protocol	Multi-protocol adapters + heavy indexing	Funded / late

For MVP, Levels 1–2 match “feels live” like Euler without insane infra cost.

13. Expanding Beyond Morpho (Adapters)

Keep Explore clean with a protocol adapter interface:

```
interface YieldAdapter { protocol: 'morpho' | 'uniswap' | 'euler' | ... listPools(chainId: number): Promise<PoolRow[]>
  getUserPositions(chainId: number, user: Address): Promise<Position[]> }
```

Adapter	Data source	Phase
Morpho	Morpho GraphQL API + contracts	Phase 1
Uniswap LP	Subgraph / position NFTs + pool state	Phase 3
Generic ERC-4626	Factory events + totalAssets / convertToAssets	Phase 3
Euler (optional multi-chain)	Euler API / subgraphs	Only if multi-network

Table: Adapter roadmap — Morpho first.

14. Costs

Item	MVP	Growing	Notes
Hosting + DB	\$20–80/mo	\$100–400/mo	Vercel/Railway/Supabase
Alchemy RPC	\$0–50/mo	\$200+/mo	Scales with portfolio polls
Morpho API	Free public	Free (cachel)	Respect rate limits via backend
Telegram / email	\$0–20/mo	\$50+/mo	Alerts
Stripe fees	2.9% + fixed	Same	On subscriptions
Domain / email	\$20–50/yr	—	One-time-ish
Legal ToS/disclaimer	\$0–1k	\$2–5k	Templates then lawyer
Smart contract audit	\$0	Only if you add deposit contracts	Stay read-only first
Cash to launch (you code)	\$500–3k	—	No salaries included
12-month opex (indie)	\$1–5k	\$5–15k	Before paid ads

Table: Cost ranges for this product specifically.

15. What Not to Do

- Do not build a full custom subgraph before trying Morpho's public API
- Do not promise “7% guaranteed” — always show base vs rewards
- Do not custody user funds or keys
- Do not list 500 random farms day one — curation is a feature

- Do not fire 50 parallel RPC calls from the browser for the Explore table
- Do not imply affiliation with Robinhood, Morpho, or Steakhouse without permission
- Do not skip geo/eligibility disclaimers (Earn and products vary by region)
- Do not add a custom fee-routing contract until revenue justifies a \$15–40k audit

16. Definition of Done (v1)

You are live when all of the following are true:

1. Explore lists Morpho vaults on RH Chain (or at least Earn + top vaults) with APY/TVL refreshing ≤ 1 minute
2. Vault page explains risk + incentive split better than Robinhood's simple Earn screen
3. Connected wallet shows my position updating within ~15 seconds
4. At least one alert type works (e.g. net APY below threshold)
5. Public URL + Stripe Pro (\$10–15/mo) for alerts and history
6. Clear disclaimers: not advice, yields variable, not FDIC/SIPC

17. Monetization

Tier	Price (example)	Includes
Free	\$0	Explore, basic vault detail, 1 wallet snapshot
Pro	\$10–20/mo	Alerts, history charts, multi-wallet, CSV export
API (later)	\$49–499/mo	Vault snapshots API for bots/creators
Affiliate (optional)	Revshare	Deep-links to venues — disclose clearly

Example: 200 Pro users × \$12 = \$2,400 MRR. Monetize insight and alerts — never “guaranteed yield.”

18. Distribution

Channel	Tactic
SEO	“How Robinhood Earn APY works”, “Morpho vault risks”, “USDG lend explained”
X / CT	Live APY breakdown posts when rates move; dashboards not memecoins
YouTube	Walkthrough: Earn under the hood + your dashboard
Reddit	Answer Earn risk questions carefully (no spam)
RH ecosystem	Email chain-developers-group@robinhood.com with live metrics
Arbitrum	Open House prizes + Foundation grants for retail transparency tooling
In-product	Shareable “vault health” cards

Table: Distribution map for this product.

19. Sources & Links

- Euler Explore (UX reference): <https://app.euler.finance/explore?network=1>
- Euler data docs (subgraphs, APIs, patterns): docs.euler.finance developers data-querying
- Morpho API / vaults GraphQL: docs.morpho.org (developers API, earn get-data tutorials)
- Morpho: “Skip Indexing, Start Building” API blog — vault APY, positions, TVL queries
- Robinhood Chain docs: <https://docs.robinhood.com/chain/>
- Robinhood Earn support / disclosures: [robinhood.com support crypto-earn](https://robinhood.com/support/crypto-earn)
- Network: Robinhood Chain mainnet chain ID 4663; Alchemy recommended RPC
- Developer contact: chain-developers-group@robinhood.com
- Arbitrum Open House: openhouse.arbitrum.io · Grants: arbitrum.foundation/grants

20. Closing Checklist

Question	Answer
Is this like Euler Explore?	Yes — vault registry + filters + portfolio
How do pools stay updated?	Backend poll Morpho API → DB every 30–60s; wallet via RPC poll / new block
Is it real-time?	Near real-time (seconds–minute), not millisecond trading
Where to start?	Morpho + Robinhood Earn on chain 4663 only
Portfolio?	Same app, tab 2 — positions from API + balanceOf
Custody?	None — read-only intelligence + optional deep-links
Audit needed for v1?	No, if you stay read-only

Question	Answer
First revenue?	Pro alerts + history via Stripe

Table: One-page summary answers.

Build the Morpho + Earn intelligence layer first. Make Explore feel as clear as Euler. Keep data fresh with a simple worker + wallet polling. Charge for alerts and portfolio history. Expand to LPs and multi-protocol only after users pay for the core job.

— End of build guide —

For product planning only. Yields, APIs, and chain integrations change. Re-verify Morpho API fields and Robinhood Earn vault addresses before production.